

8086 BUS Timing

- mips চি প্রস্তুত instruction চি 5 টি clock cycle লাগে।
যদি 8086 চি at most 6 টি clock cycles লাগে।
- এখন আমরা আরও চি general micro-processor study করবো।

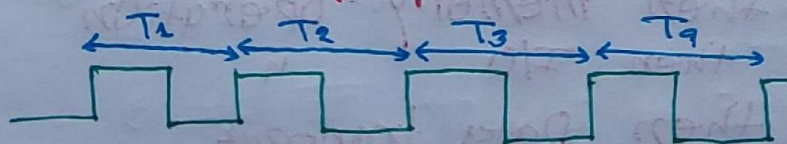
Machine cycles:

A machine (Bus) cycles consist of at least four clock cycles, called T-states.

CPU যখন memory বা I/O device চি স্মার communicate করে তখন চি প্রস্তুত operation হয় আর machine cycle বলে।

একটি machine cycle = ৪ টি T-states.

Total 4 টি clock cycles চি 8086 চি memory/I/O and read/write operation complete করে।

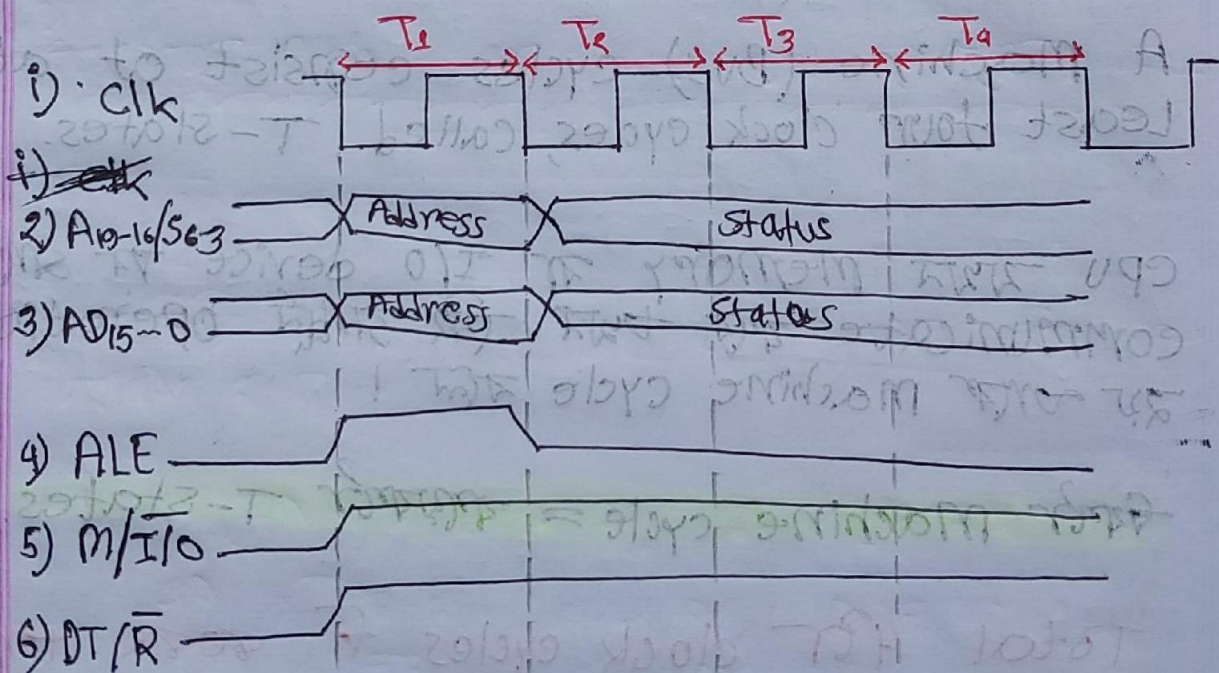


T₁

Address Bus related ~~बाद~~ बाय, mainly CPU address Bus ~~के~~ address है।

SO T₁ ~~के~~ active रह / बाय बाय ~~है~~ है
Pin सूला, सिग्नल clock(Clk), A₁₀₋₁₆/S₂₋₃
and AD₁₅₋₀ is common for all ~~question~~
सिग्नल read ~~के~~ AD₁₅₋₀ का सिग्नल float रहता
लाता।

बाय ~~उपर~~ start रह T₁ सिग्नल ~~बाय~~ ~~मिस~~ depend
बाय value change रह।



if ALE = 1 then, AD₀₋₁₅ is Address

M/ $\bar{I/O}$ = 1 then memory operation
= 0 then I/O

DT/ $\bar{R}$ = 1 then Data transmit
= 0 " " Read

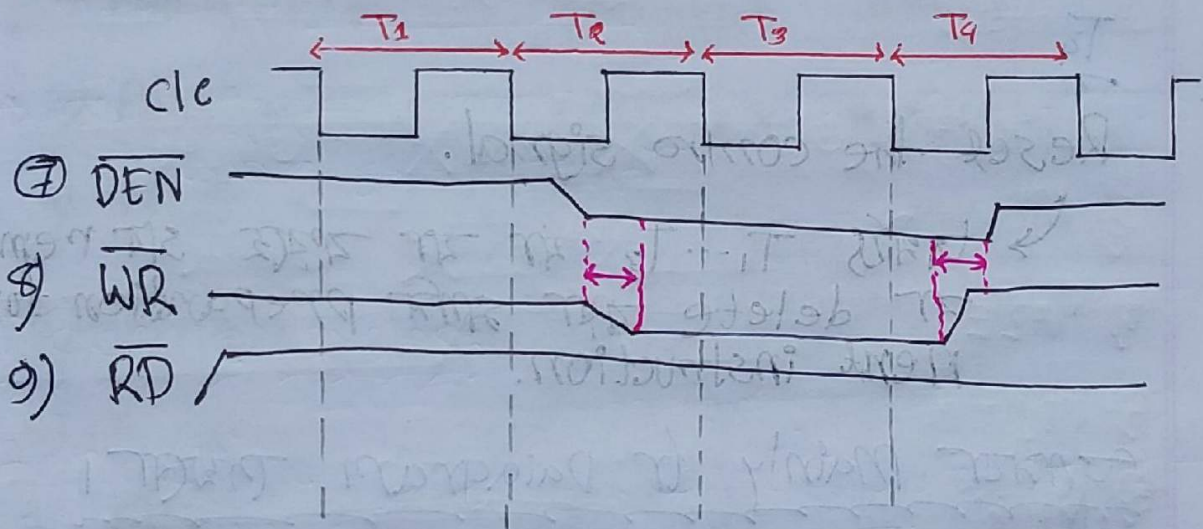


સિગ્નલ માટે exact value નક્કી કરવામાં આવે છે (Low or high)

T₂:

prepare data bus

→ Address નાં આધારે data બસ પર, system તૈયાર થાય છે.



અહીં અવકાશ બેસાડવામાં આવે છે કે DEN, WR જે સિગ્નલ આપે છે તે સ્થિતિમાં છે.

સાધન

→ જો DEN Active હોય તો system તૈયાર થાય છે અને આમ Ready થઈ Data મોકલે (CPU $\xrightarrow{\text{Data}}$ memory)

NB અહીં WR સિગ્નલ આપે છે જે RD Off થાય છે.

0 = 0/10 10 1 = 0/10 00

T₃

Data transfer st (include wait state)

micro-processor ~~आता~~ fast सिद्ध I/O device slow छि wait कर लाह sometimes

T₄

Reset the contro signal.

→ प्रत्येक T₁...T₃ या या चक्रहरू अर remove वा delete कर झनि preparation for next instruction.

अन्तर्गत mainly for Daigram लख्नु ।

i) memory write

अवस्थिति memory write छि अवस्थिति दिनुना बखि रह ।

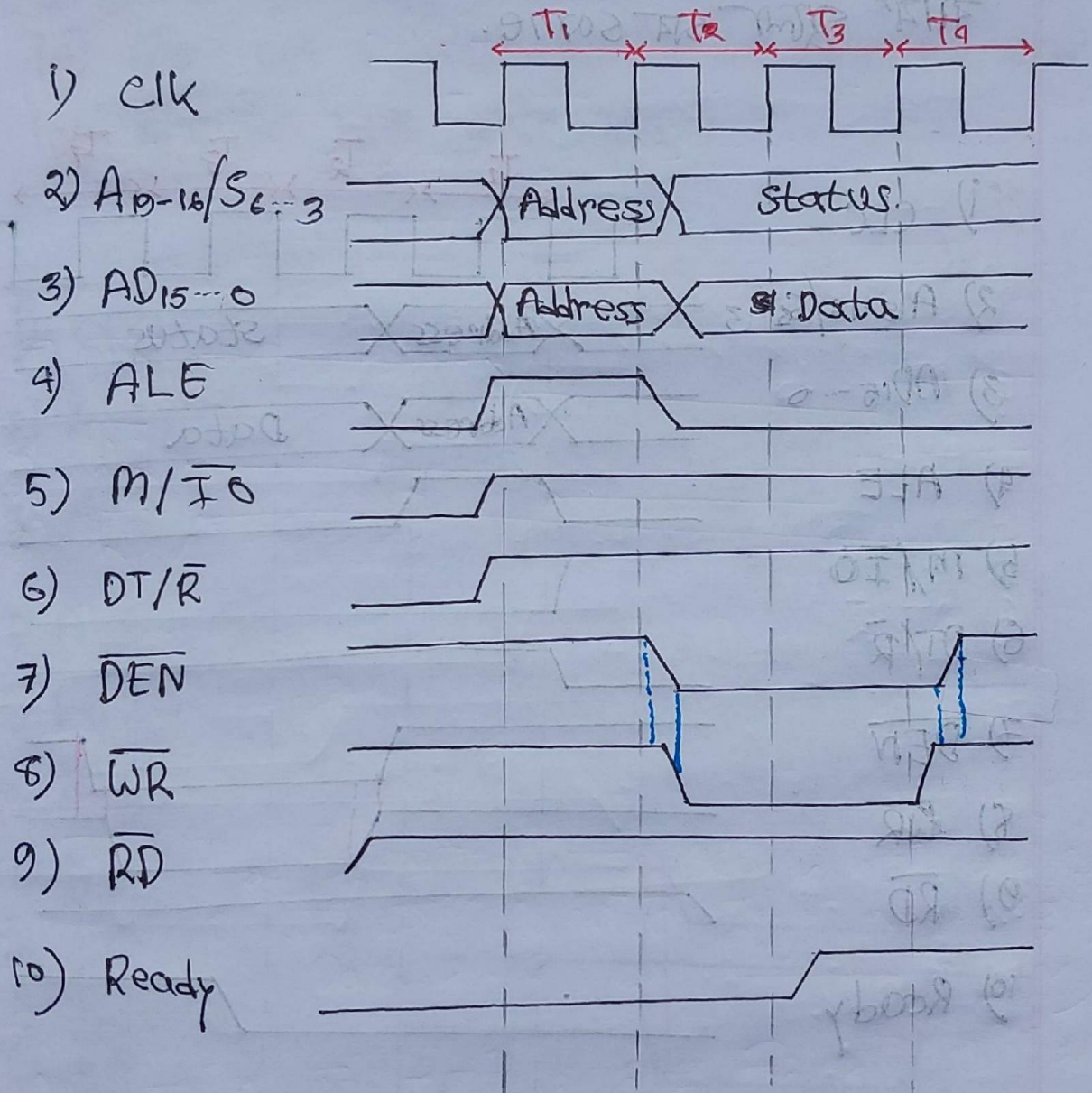
i) $M/\overline{IO} \rightarrow$ memory वा high बखि रह
so $\overline{M/\overline{IO}}$ रह memory = 0 झनि high
or $M/\overline{IO} \parallel$ memory = 1 " "

ii) memory/IO write झनि Data ~~trans~~ transmit

so, $DT/\overline{R} \Rightarrow 1$ or $\overline{DT/R} = 0$

iii) $\overline{WR}$ → મરિયુ સાર જાણે 50, 0 અને active, 1 અને inactive.

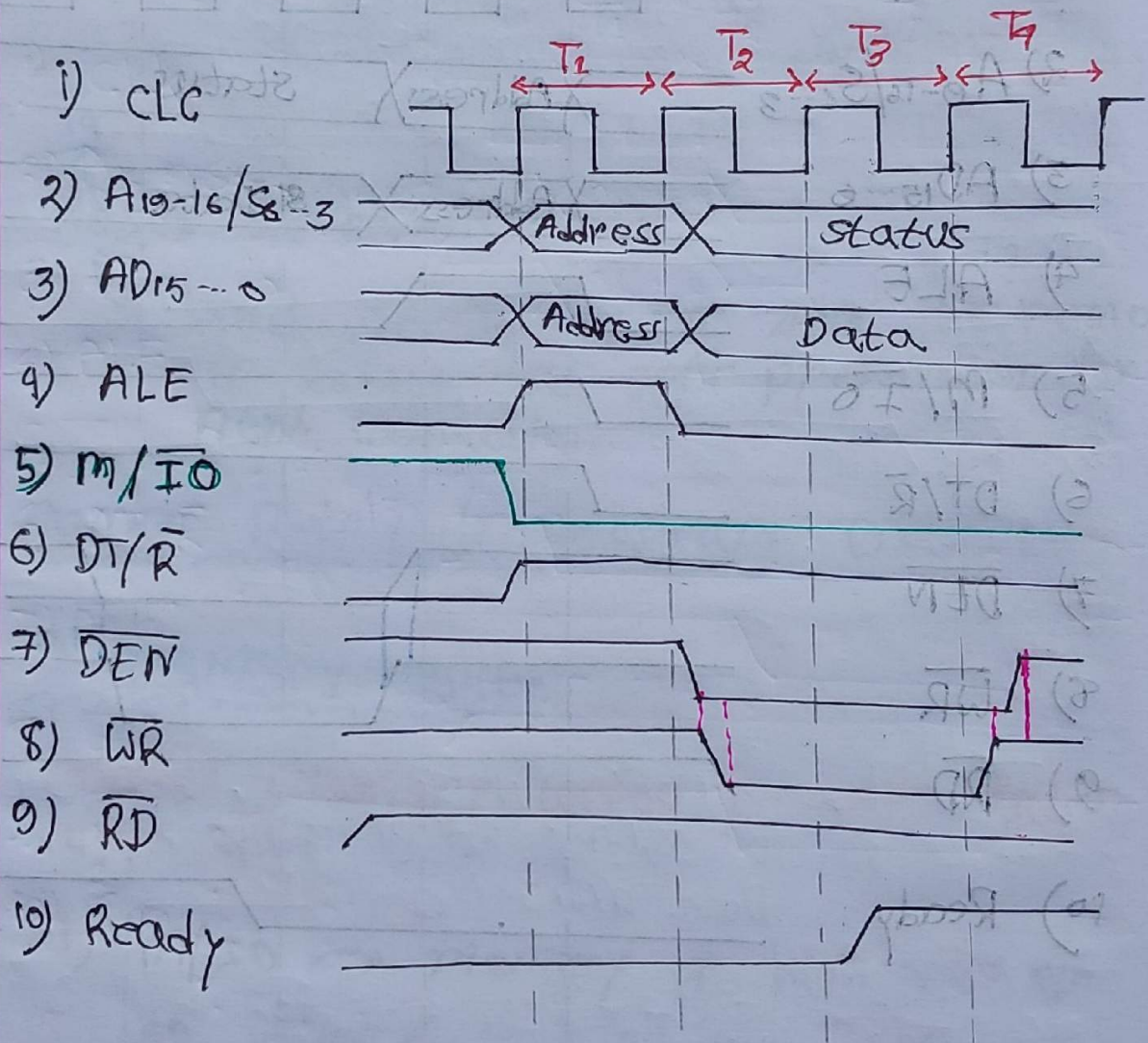
iv) $\overline{RD}$ → જ્યારે $\overline{RD}$ off થાય ત્યારે 50 1 અને off



in write → start $\overline{DEN} \rightarrow \overline{WR}$
 → end $\overline{DEN} \rightarrow \overline{WR}$

ii) I/O write

যদিও memory write এর only M/I/O pin এর value change থাকবে।
বাকি সবার সব same.

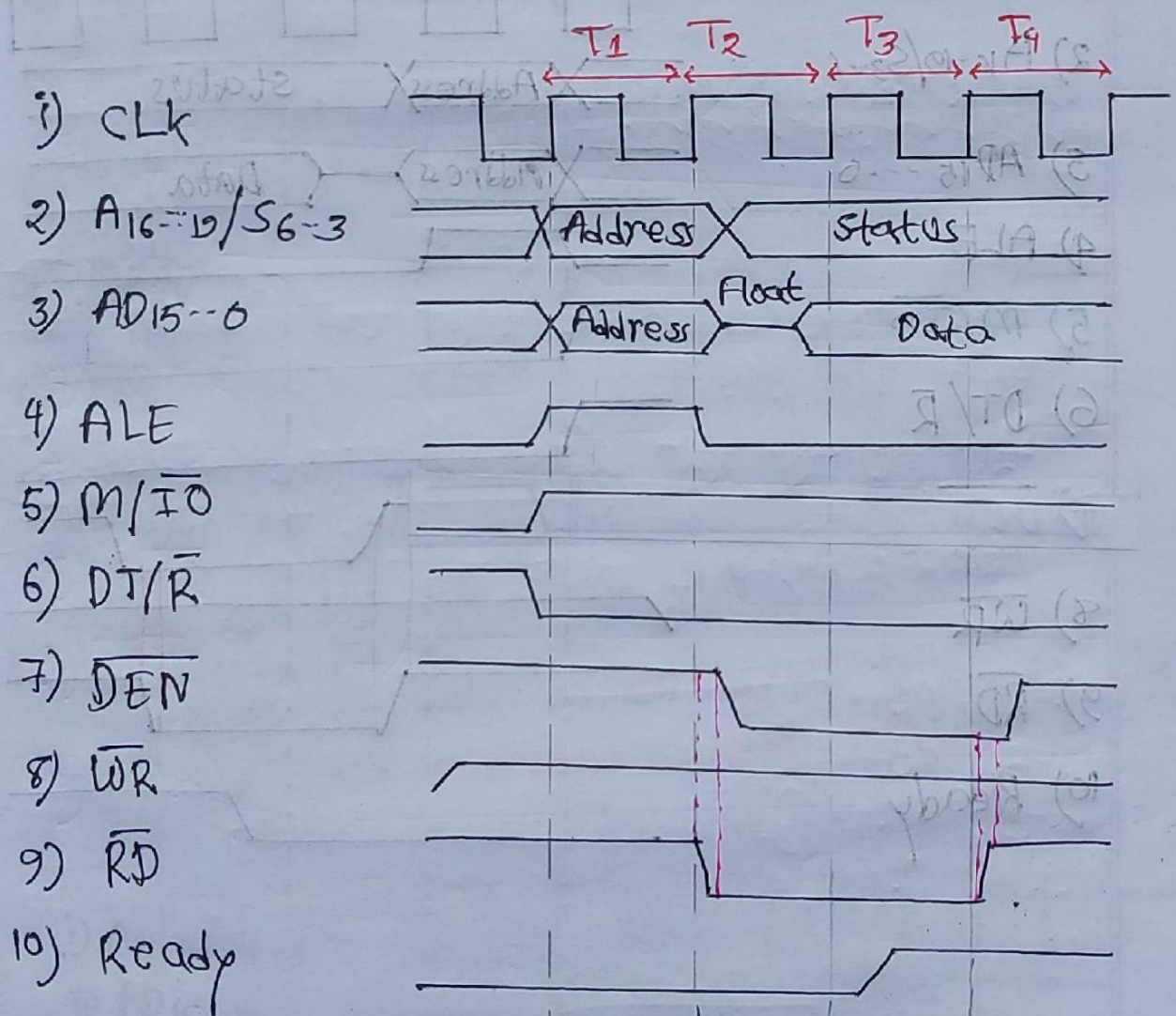


iii) Write operation DEN, WR, RD

Float चककर काय Data CPU ले नरि
जरे DEN on चककर सगल सगल Data
वाक नरि !

iii) memory Read

- i) चककर M/I $\bar{O}$ ले memory high चककर
- ii) DT/R ले R ले 1 चककर रर 50 R DT/R=0
- iii) WR ऑफ चककर रर, 50 WR=1 सगल ऑफ
- iv) RD ले high चककर रर ।

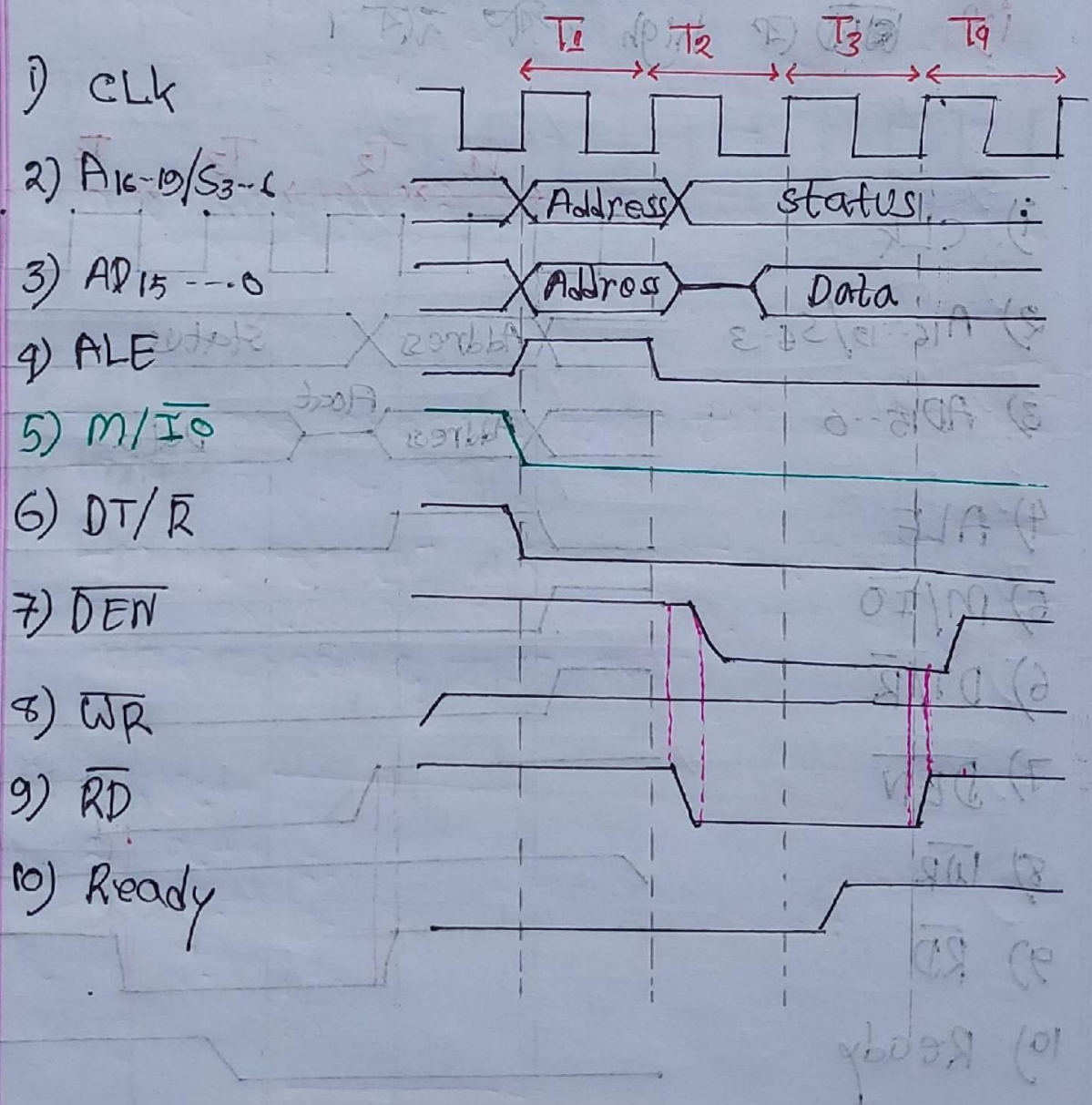


Read चककर सगल AD₁₅₋₀ ले T₂ ले चककर
Float सगल सगल DEN and RD on नरि रररर

in Read $\rightarrow$ start : RD > DEN
end : RD > DEN

iv) I/O Read:

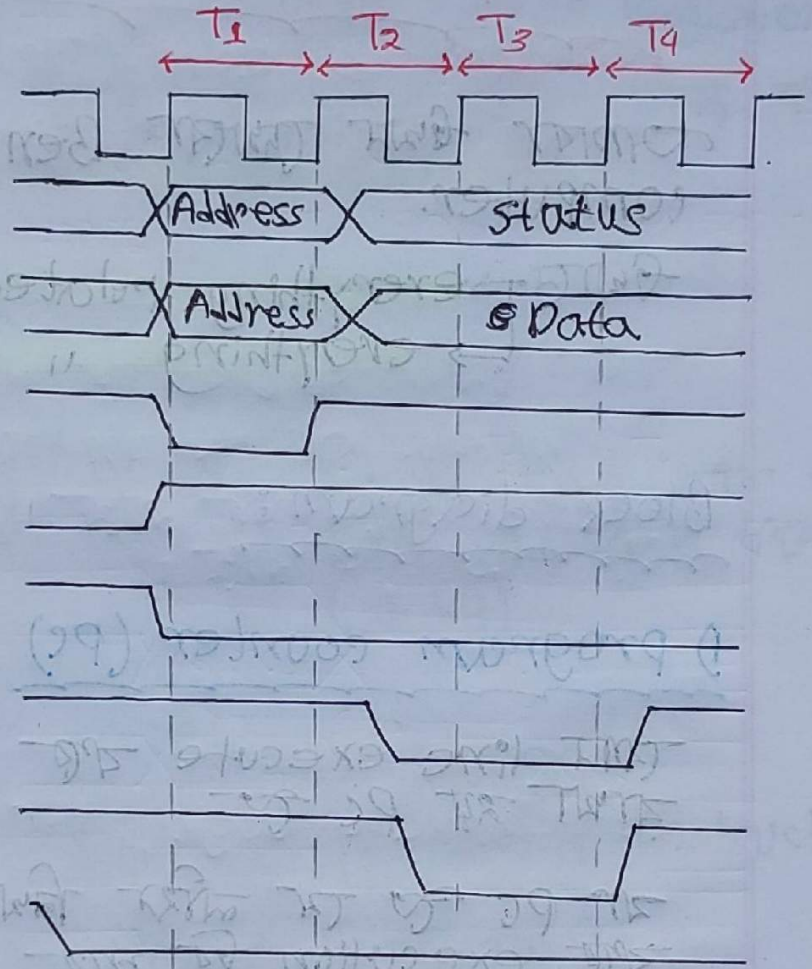
যদি memory Read হ'লে Only M/I/O pin
এর value change হওয়া থাকবে।



I/O Write

summer 25 Q3(b)

- 1) CLK
- 2) A₁₆₋₁₉/S₃₋₆
- 3) AD₁₅₋₀
- 4) ALE
- 5) M/I/O
- 6) DT/R
- 7) DEN
- 8) WR
- 9) RD



fall 25 Q2(b)

mainly error due to data bus and correction error

- i) A₁₆₋₁₉/S₃₋₆
- ii) AD₁₅₋₀
- iii) DT/R
- iv) DEN
- v) M/I/O
- vi) RD

